Root parameters and fixed-space obstructions

A source-attributed recovery for the Markovskaya–Nuzhin classifications

This common paper supplies elementary proofs of the root-recognition and fixed-space tools used in the accompanying linear-group application. The results are classical; no priority claim is made. In particular, the root recognition specializes Levchuk's theorem, and the fixed-space inequality is Scott's. All fields in the root-recognition statements are finite. Write $t_{ij}(a) = I + aE_{ij}$, and use $x^y = yxy^{-1}$ and $[x, y] = xyx^{-1}y^{-1}$.

Lemma 1 (Root propagation)**.** *Let $M \le \mathrm{GL}_n(F)$ and let a directed graph on $\{1, \dots, n\}$ have an arc $i \to j$ when M contains some $t_{ij}(a)$ with $a \neq 0$. If this graph is strongly connected, M meets every root subgroup nontrivially. In particular a monomial matrix with permutation part an n-cycle, together with a nonidentity root element between cyclically consecutive positions, has this property, for either orientation.*

Proof. For distinct i, j, k, elementary matrix multiplication gives $[t_{ij}(a), t_{jk}(b)] = t_{ik}(ab)$. A simple directed path therefore supplies its endpoint root by induction; its vertices are distinct, so every use of this formula is legitimate. Conjugation by a monomial matrix transports a root to the root on the permuted ordered pair and multiplies its parameter by a nonzero scalar. The conjugates of the indicated adjacent root give a directed Hamiltonian cycle. The assertion for $n = 2$ needs only these two conjugates, not a three-index commutator. □

Theorem 2 (Finite root-parameter recognition)**.** *Let $n \ge 3$ and $M \le \mathrm{GL}_n(F_q)$ meet every root subgroup nontrivially. Put $A_{ij} = \{a : t_{ij}(a) \in M\}$ and $R = \langle t_{ij}(A_{ij}) : i \neq j \rangle$. There are a subfield $K \le F_q$ and an invertible diagonal matrix D such that $DRD^{-1} = \mathrm{SL}_n(K)$. This does not assert that DMD^{-1} equals $\mathrm{SL}_n(K)$. Consequently every determinant-one diagonal sign matrix belongs to R, and R contains a monomial lift of every coordinate permutation. If $a, b \in A_{ij}$ with $b \neq 0$, then $a/b \in K$; if $a \in A_{ij}$ and $b \in A_{ji}$, then $ab \in K$.*

Proof. Each A_{ij} is a nonzero finite additive group and $A_{ij}A_{jk} \subseteq A_{ik}$ for distinct indices. Multiplication by a fixed nonzero element gives $|A_{ij}| \le |A_{ik}|$; interchanging j, k gives equality. Thus the size is constant in each row. Similarly left multiplication shows it is constant in each column. For $n \ge 3$ the off-diagonal positions are connected by row and column steps, so all sizes are the same number m. Every preceding product with one fixed nonzero factor is therefore an equality.

Choose $a_i \in A_{i1} \setminus \{0\}$ for $i > 1$ and conjugate by $D = \mathrm{diag}(1, a_2^{-1}, \dots, a_n^{-1})$. Write B_{ij} for the new parameter groups. Then $1 \in B_{i1}$. For distinct $i, j > 1$, $B_{ij} \subseteq B_{i1}$, by multiplying by $1 \in B_{j1}$, and equality follows from the sizes. Also $B_{1j} \subseteq B_{ij}$ by multiplying by $1 \in B_{i1}$, so

$$B_{ij} = B_{i1} = B_{1j} =: K_i \qquad (i, j > 1,\ i \neq j).$$

The product $B_{i1}B_{1j} \subseteq B_{ij}$ says $K_iK_i \subseteq K_i$. Thus K_i is a finite subring of the field containing 1, hence a field: multiplication by a nonzero element is injective and surjective on K_i. Finally $B_{ij}B_{j1} \subseteq B_{i1}$ gives $K_iK_j \subseteq K_i$, hence $K_j \subseteq K_i$ and equality by sizes. There is at least one such j for each i because $n \ge 3$. All the B_{ij} are consequently the same field K. Elementary row elimination proves that their transvections generate $\mathrm{SL}_n(K)$.

Diagonal conjugation preserves diagonal matrices and monomial permutation parts. Determinant-one sign matrices lie in $\mathrm{SL}_n(K)$, as do monomial lifts of transpositions (use the 2×2 block $\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$). The ratio assertion and the opposite-root product assertion follow because the diagonal conjugating factors cancel. □

Corollary 3 (Primitive-square test and isolation)**.** *Suppose q is odd and u generates F_q^*. Then $F_p(u^2) = F_q$. Accordingly, in Theorem 2, either two same-root parameters with ratio u^2, or opposite-root parameters with product a nonzero prime-field multiple of u^2, force $R = \mathrm{SL}_n(F_q)$. For $n \geq 4$, if i, j, k are distinct and $t_{ij}(a)t_{kj}(b) \in M$, each of these two root factors can be isolated in M when q is odd.*

Proof. If u^2 lay in a proper subfield of size p^d, then $p^d \leq \sqrt{q}$ and $(q-1)/2 \leq p^d - 1$, impossible since $q + 1 > 2\sqrt{q}$. A prime field has no proper subfield, covering that case as well. For isolation choose an index ℓ different from i, j, k. The diagonal matrix with minus signs at k, ℓ belongs to R. Its commutator with the displayed product is $t_{kj}(-2b)$. An integer power multiplies this coefficient by $(-2)^{-1} \in F_p$, yielding $t_{kj}(b)$. Divide it out to obtain the other factor. This uses the previously established R; it is not a substitute for proving that all roots were reached. □

Lemma 4 (Symmetric square)**.** *For a field F of characteristic different from 2 and $n \geq 2$, the natural action of $\mathrm{GL}_n(F)$ on $\mathrm{Sym}^2(F^n)$ is irreducible and has kernel $\{I, -I\}$.*

Proof. Use the basis $e_i^2, e_i e_j$ $(i < j)$. The diagonal sign group has distinct characters on the off-diagonal basis vectors, and the trivial character on the span of the squares. Averaging against its characters is permitted, since its order is a power of 2. Hence an invariant subspace with any nonzero off-diagonal coefficient contains an individual $e_i e_j$. Permutation matrices give all such vectors. Replacing e_i by $e_i + e_j$ then yields e_j^2 by subtraction, and all squares follow. If instead a nonzero vector has only square coefficients, choose $a_i \neq 0$ in $\sum a_i e_i^2$ and replace e_i by $e_i + e_j$; the new off-diagonal coefficient is $2a_i \neq 0$. Thus every nonzero invariant subspace is full. If g acts trivially, $(ge_i)^2 = e_i^2$ forces $ge_i = \epsilon_i e_i$, $\epsilon_i \in \{1, -1\}$. The mixed products force all ϵ_i equal. Both scalar signs do act trivially. □

Theorem 5 (Scott's inequality, with proof)**.** *Suppose $g_1 \cdots g_m = 1$ generate an irreducible subgroup of $\mathrm{GL}(V)$, where $\dim V \geq 2$. Then*

$$\sum_i \operatorname{codim} \operatorname{Fix}(g_i) \geq 2 \dim V.$$

Proof. Put $p_i = g_1 \cdots g_{i-1}$ and $W = \bigoplus_i \operatorname{im}(g_i - I)$. The map $\Phi : W \to V$, $(w_i) \mapsto \sum_i p_i w_i$, is onto. Indeed a covector annihilating its image annihilates $g_1 - I$ and hence is fixed by g_1; inductively it is fixed by every g_i. The dual has no nonzero common fixed vector, since its annihilator would be a proper invariant hyperplane in V. The injective map $v \mapsto ((g_i - I)v)_i$ lands in $\ker \Phi$, by the telescoping identity $\sum_i p_i(g_i - I) = g_1 \cdots g_m - I = 0$. Its kernel is the common fixed space, zero by irreducibility and $\dim V > 1$. Therefore $\dim W \geq \dim V + \dim V$. □

Corollary 6. $\mathrm{GL}_4(3)$ *cannot be generated by three involutions two of which commute.*

Proof. If a, b, c were such generators with $ab = ba$, apply the preceding results to the ten-dimensional symmetric square and the five factors a, b, ab, c, c, whose product is 1. An involution with an r-dimensional negative eigenspace has symmetric-square fixed-space codimension $r(4 - r) \leq 4$. Scott's lower bound is 20, so all five codimensions must be 4, in particular those of a, b, c. They all have $r = 2$, determinant 1, and cannot generate the full general linear group. □

Attribution and boundary. The companion source is I. A. Markovskaya and Ya. N. Nuzhin, *On generation of the groups* $\mathrm{GL}_n(q)$ *and* $\mathrm{PGL}_n(q)$ *by three involutions, two of which commute*, Trudy Inst. Mat. Mekh. UrO RAN 31(4) (2025), 247–259, DOI 10.21538/0134-4889-2025-31-4-fon-03. Its Lemmas 6–9 invoke Nuzhin's symmetric-square theorem, Scott's *Matrices and cohomology* (1977), and Levchuk's root-generation theorem (1983). The elementary proofs here replace those particular invocations, not their general published theorems. No finite search, symbolic software, or formal certificate is a premise of this paper.